Calum Murray

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Profile

I am currently a postdoctoral researcher at CosmoStat (CEA Saclay), working on observational cosmology. My research focuses on understanding how galaxies populate the cosmic web by combining weak gravitational lensing observations with detailed characterization of observational systematics. I develop physically motivated models and forward-modeling strategies validated on real survey data from Euclid, UNIONS, and DESI, with particular focus on cluster cosmology and characterizing the galaxy-halo connection through both galaxy clustering and intrinsic alignments. I hold leadership positions in the Euclid collaboration as co-lead of the cluster catalogue work package and the COMB-CL processing function for cluster weak lensing measurements.

Research experiences

01/09/2024 – Present: Postdoctoral researcher at CosmoStat, CEA Saclay, France
Advisors: Dr. Martin Kilbinger & Dr. Samuel Farrens

01/09/2022 – 01/09/2024: Postdoctoral researcher at AstroParticle and Cosmology laboratory (APC), Université de Paris, France
Advisor: Prof. James G. Bartlett

01/11/2020 – 01/09/2022: Postdoctoral researcher at Laboratory of Subatomic Physics and Cosmology (LPSC), Université Grenoble Alpes, France
Advisor: Dr. Céline Combet

Education and key qualifications

30/10/2020: PhD, Astrophysics, Université Paris Cité, France
Thesis: *Galaxy cluster cosmology with the Euclid Dark Energy Survey*
Supervisor: Prof. James G. Bartlett

2017: Master of Mathematical Physics (MPhys), University of Edinburgh, UK
Thesis: *The colours of galaxies*
Supervisor: Prof. Robert Mann

Leadership roles in major collaborations

Euclid Consortium (2017 – Present)

- **Co-lead** of the Galaxy Clusters work package on catalogue selection and definition (since 2024)
- **Co-lead** of COMB-CL processing function for cluster weak lensing measurements (improved code speed by factor of 50 for large datasets) (since October 2025)
- **Lead author** on key paper analysing covariances between observed cluster properties using the 5,150 square degree Euclid Flagship simulation, in preparation.

UNIONS – Ultraviolet Near Infrared Optical Northern Survey (2024 – Present)

- Member, contributing to cosmic shear analysis and intrinsic alignment measurements using UNIONS galaxy shape measurements combined with DESI spectroscopic data

LSST Dark Energy Science Collaboration (2020 – Present)

- Member, contributed two-halo term computation and elliptical halo validation to DESC cluster weak lensing processing function CLMM

Wide-field Spectroscopic Telescope (WST)

- Member of WST Science Team

Statistical and methodological expertise

- Weak gravitational lensing: shear and magnification measurements, cluster mass estimation
- Development of statistical techniques: Wiener filtering, two-dimensional shear field analysis
- Forward modeling and simulation-based inference approaches
- Cross-correlation analyses for tests of gravity and cosmological systematics
- Extended GPU correlation code (cucount, https://github.com/adematti/cucount) for arbitrary spin-correlations, enabling rapid computation of shape-position and higher-order correlations (100x speedup)
- Cluster abundance cosmology: likelihood development, super-sample covariance
- Large-scale survey data validation and systematic characterization
- Analysis of photometric surveys (Euclid, UNIONS, HSC) and spectroscopic surveys (DESI)
- Production of mock observations and catalogue-level forward models

Supervising & Teaching

PhD Supervision

- Antonin Corinaldi (Sept 2025 – present), intrinsic alignments with UNIONS and DESI, co-supervised with Dr. M. Kilbinger and Dr. S. Codis
- Constantin Payerne (co-supervised during his PhD), cluster abundance cosmology, resulted in 2 publications
- Raphaël Kou (co-supervised during his PhD), tests of modified gravity, resulted in 1 publication

Master's Students (5 total)

- Antonin Corinaldi (Jan–Sept 2025), intrinsic alignment with UNIONS and DESI
- Huy Nguyen Phu (Apr–Sept 2024), galactic acceleration using quasar proper motions
- Théo Courty (May–July 2023), intrinsic alignments of galaxies
- Mahmoud Osman (Apr–June 2022), optical mass proxies for galaxy clusters
- Margaux Scavini (May–June 2019), strong gravitational lensing

Teaching

- August 2025: Lecturer, Euclid Summer School. Delivered 6-hour graduate-level course on observational aspects of weak gravitational lensing
- June 2025: Lecturer, COLOURS Workshop. Developed and taught workshop on analyzing Euclid's first data release

Contributions to the research community

Scientific life

- Since 2025: Peer reviewer for American Astronomical Society (AAS) journals
- 2022–2024: Organizer of weekly APC cluster journal club
- 2018–2020: Organizer of biweekly APC cosmology journal club
- 2021: Local Organizing Committee, Euclid France meeting, LPSC Grenoble
- 2019: Organizing Committee, Rencontres des Jeunes Physicien.ne.s, Collège de France, Paris

Outreach activities

- Since Nov 2023: Co-organizer of Astronomy on Tap, Paris. Organized and hosted 12 public lectures making astronomy accessible to general audiences
- Nov 2022 & Dec 2024: Rencontres du Ciel et de l'Espace, Paris. Volunteered at Euclid and Rubin stands explaining cosmological surveys to the public

Selected publications (25 total, including 8 first-author, add number of 2nd author and 4 as co-supervisor)

First-author publications

C. Murray, C. Combet, C. Payerne, M. Ricci (2025). Filtering out large-scale noise for cluster weak lensing mass estimation. *Astronomy & Astrophysics*, 697, A141

C. Murray, J.G. Bartlett, E. Artis, J.B. Melin (2022). Measuring weak lensing masses on individual clusters (catalogue of 3000+ clusters). *Monthly Notices of the Royal Astronomical Society*, 512(4), 4785-4791

C. Murray (2022). The effects of lensing by local structures on the dipole of radio source counts. *Monthly Notices of the Royal Astronomical Society*, 510(2), 3098-3101

C. Murray, J.G. Bartlett, R. Kou (submitted to A&A). Gravitational lensing by matter currents. arXiv:2510.19798

C. Murray, M. Kilbinger (under internal review within UNIONS). Observatational evidence for the origin of intrinsic alignments of galaxies.

C. Murray (in preparation). Two-dimensional shear field estimation of cluster masses, concentration and halo-miscentering

Euclid Collaboration: **C. Murray**, S. Bhargava, et al. (in preparation). Euclid preparation: Covariances between observed cluster properties with the Euclid Flagship simulation

Key collaborative publications (co-supervised students)

C. Payerne, **C. Murray**, C. Combet, C. Doux, M. Penna-Lima (2024). Cluster abundance cosmology: Including super-sample covariance in the unbinned likelihood. Submitted to A&A, arXiv:2401.10024

C. Payerne, **C. Murray**, C. Combet, C. Doux, A. Fumagalli, M. Penna-Lima (2023). Testing the accuracy of likelihoods for cluster abundance cosmology. *Monthly Notices of the Royal Astronomical Society*, 520(4), 6223-6236

C. Payerne, **C. Murray**, et al. (in preparation). Simulation-based inference framework for optical cluster cosmology incorporating selection effects and galaxy population systematics

R. Kou, **C. Murray**, J.G. Bartlett (2024). Constraining f(R) gravity with cross-correlation of galaxies and cosmic microwave background lensing. arXiv:2311.09936

Other significant collaborative publications

E. Artis, J.B. Melin, J.G. Bartlett, **C. Murray** (2021). Impact of the calibration of the halo mass function on galaxy cluster number count cosmology. *Astronomy & Astrophysics*, 649, A47

Euclid Collaboration publications

Euclid Collaboration: S. Bhargava, C. Benoist, A.H. Gonzalez, ..., **C. Murray**, et al. (2025). Euclid Quick Data Release (Q1): First detections from the galaxy cluster workflow. arXiv:2503.19196

Plus 7 additional published Euclid papers, 1 UNIONS paper submitted, 2 DESC papers under internal review

Add UNIONS papers,

Fabian 2024, 3 UNIONS in internal review

Invited talks & conference presentations

Selected invited seminars

- July 2025: Le comptage d'amas de galaxies dans notre Univers avec Euclid, 27ème Congrès Général de la Société Française de Physique, Troyes, France
- January 2025: Constraining f(R) gravity with cross-correlation of galaxies and cosmic microwave background lensing, Max Planck Institute for Extraterrestrial Physics, Munich, Germany
- April 2024: Beyond cosmic variance: galaxy cluster weak lensing, Royal Observatory of Edinburgh, United Kingdom
- March 2024: Robust estimation of the quasar dipole, Laboratoire Univers et Particules de Montpellier, France
- February 2023: Gravitational lensing by matter currents, CPPM, Marseille, France
- January 2021: Individual weak lensing masses with HSC, LPSC, Grenoble, France
- January 2020: Individual weak lensing masses with HSC, CosmoStat, CEA Saclay, France
- December 2018: Individual weak lensing masses for Euclid, Observatoire de Paris, Meudon, France

Conference & collaboration presentations

8 international conferences (including Rencontres de Moriond, Tracing Cosmic Evolution with Clusters of Galaxies Sexten, Progress on Old and New Themes in Cosmology Avignon) · 25 Euclid collaboration meetings (Euclid France, general Consortium meetings, Galaxy Clusters Science Working Group, Theory Science Working Group) · 8 LSST Dark Energy Science Collaboration meetings

References

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